Mean Squared Error(MSE)

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weight update:

Let y = Prediction

Let A = Actual

$MSE = (y-A)^2$ To reduce MSE, we need to adjust w and bias

Find $dMSE/dw$ and $dMSE/dbias$

$dMSE/dw = dMSE/dy \cdot dy/dw + dMSE/dA \cdot dA/dw =$

$dMSE/dy \cdot dy/dw + 0$ (Because Actual is independent from w)

$dMSE/dy \cdot dy/dw = 2(y-A) \cdot dy/dw(y-A) = 2(y-A)\, dy/dw(y)$

$dy/dw_n = x_n$, then $dy/dw(y) = X$

$X^T \cdot 2(y-A) \rightarrow$ Dot product] Dense Layer does this part. Simply return $2(y-A)$

Bias update:

Nearly identical reasoning to weight updates:

$dy/db_n = 1$, then bias updates are simply

$2(y-A)$ per bias